



Import libraries

```
In [1]: import pandas as pd
import numpy as np

import matplotlib.pyplot as plt
import seaborn as sns
```

```
In [2]: data = pd.read_csv('final_vg.csv')
data.head()
```

```
Out[2]:
```

	Rank	Name	Platform	Year	Genre	Publisher	NA_Sales	EU_Sal
0	2061	1942	NES	1985.0	Shooter	Capcom	4.569217	3.0338
1	9137	[Shin Chan Flipa en colores!	DS	2007.0	Platform	505 Games	2.076955	1.4934
2	14279	.hack: Sekai no Mukou ni + Versus	PS3	2012.0	Action	Namco Bandai Games	1.145709	1.7623
3	8359	.hack//G.U. Vol.1//Rebirth	PS2	2006.0	Role- Playing	Namco Bandai Games	2.031986	1.3898
4	7109	.hack//G.U. Vol.2//Reminisce	PS2	2006.0	Role- Playing	Namco Bandai Games	2.792725	2.5920

```
In [3]: top3_pub = data['Publisher'].value_counts().index[:3]
top3_gen = data['Genre'].value_counts().index[:3]
top3_plat = data['Platform'].value_counts().index[:3]
top3_data = data.loc[(data["Publisher"].isin(top3_pub)) & (data["Platform"].is
top3_data.head()
```

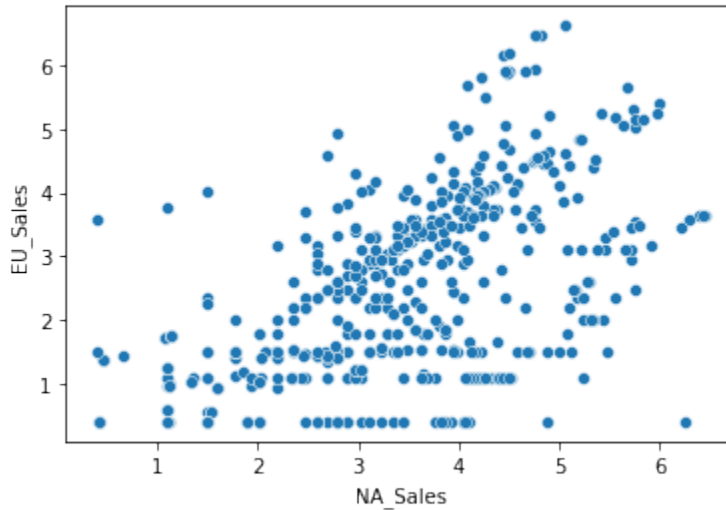
```
Out[3]:
```

	Rank	Name	Platform	Year	Genre	Publisher	NA_Sales	EU_Sales	
2	14279	.hack: Sekai no Mukou ni + Versus	PS3	2012.0	Action	Namco Bandai Games	1.145709	1.762339	1
13	2742	[Prototype 2]	PS3	2012.0	Action	Activision	3.978349	3.727034	0
16	1604	[Prototype]	PS3	2009.0	Action	Activision	4.569217	4.108402	1
19	1741	007: Quantum of Solace	PS3	2008.0	Action	Activision	4.156030	4.346074	1
21	4501	007: Quantum of Solace	PS2	2008.0	Action	Activision	3.228043	2.738800	2

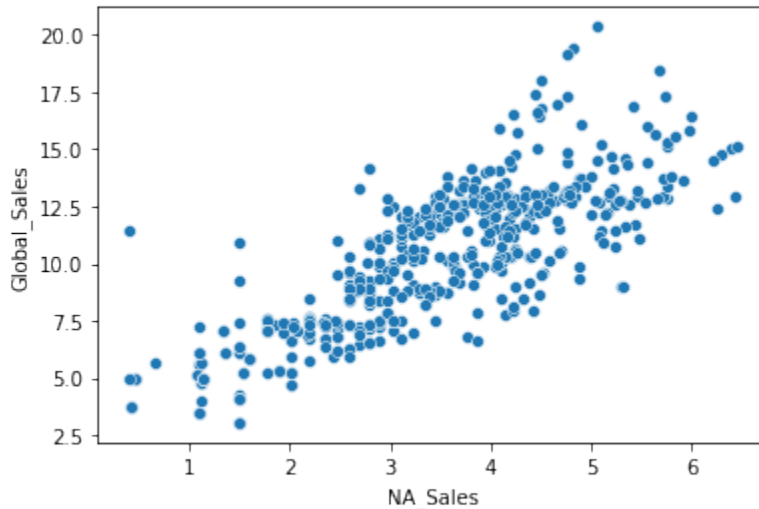
In []:

Subplots

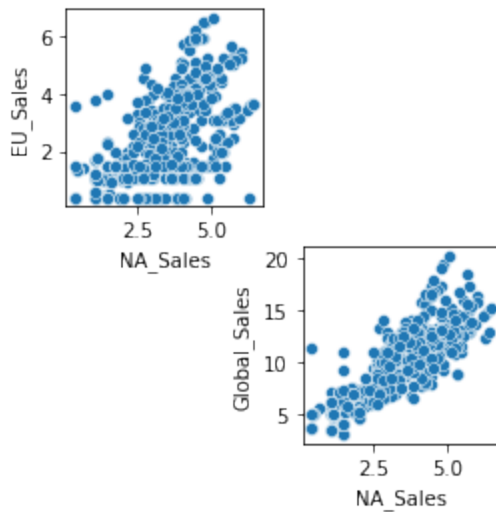
```
In [4]: sns.scatterplot(data=top3_data, x = 'NA_Sales', y = 'EU_Sales')
plt.show()
```



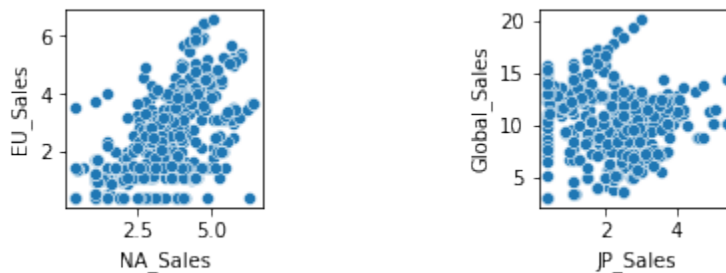
```
In [5]: sns.scatterplot(data=top3_data, x = 'NA_Sales', y = 'Global_Sales')
plt.show()
```



```
In [7]: plt.subplot(2, 3, 1) # 2 rows and 3 columns, 1 -> slot/position
sns.scatterplot(data=top3_data, x = 'NA_Sales', y = 'EU_Sales')
plt.subplot(2, 3, 5)
sns.scatterplot(data=top3_data, x = 'NA_Sales', y = 'Global_Sales')
plt.show()
```

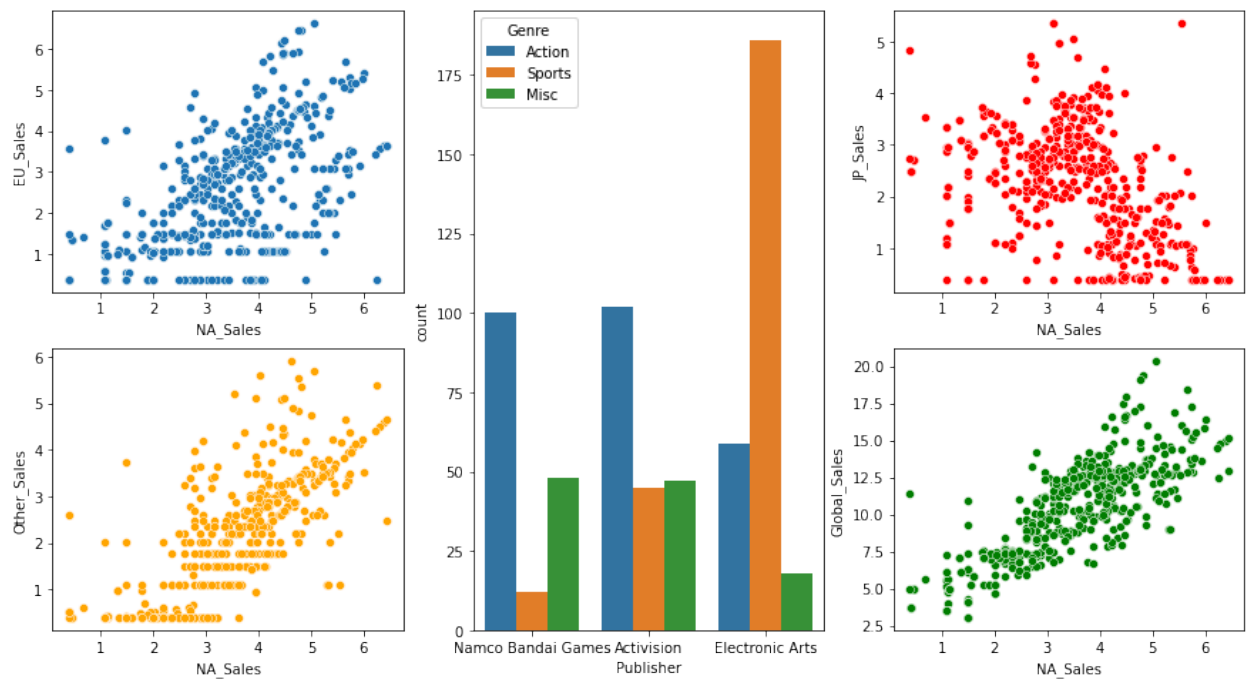


```
In [17]: plt.subplot(2, 3, 1) # 2 rows and 3 columns, 1 -> slot/position
sns.scatterplot(data=top3_data, x = 'NA_Sales', y = 'EU_Sales')
plt.subplot(2, 3, 3)
sns.scatterplot(data=top3_data, x = 'JP_Sales', y = 'Global_Sales')
plt.show()
```



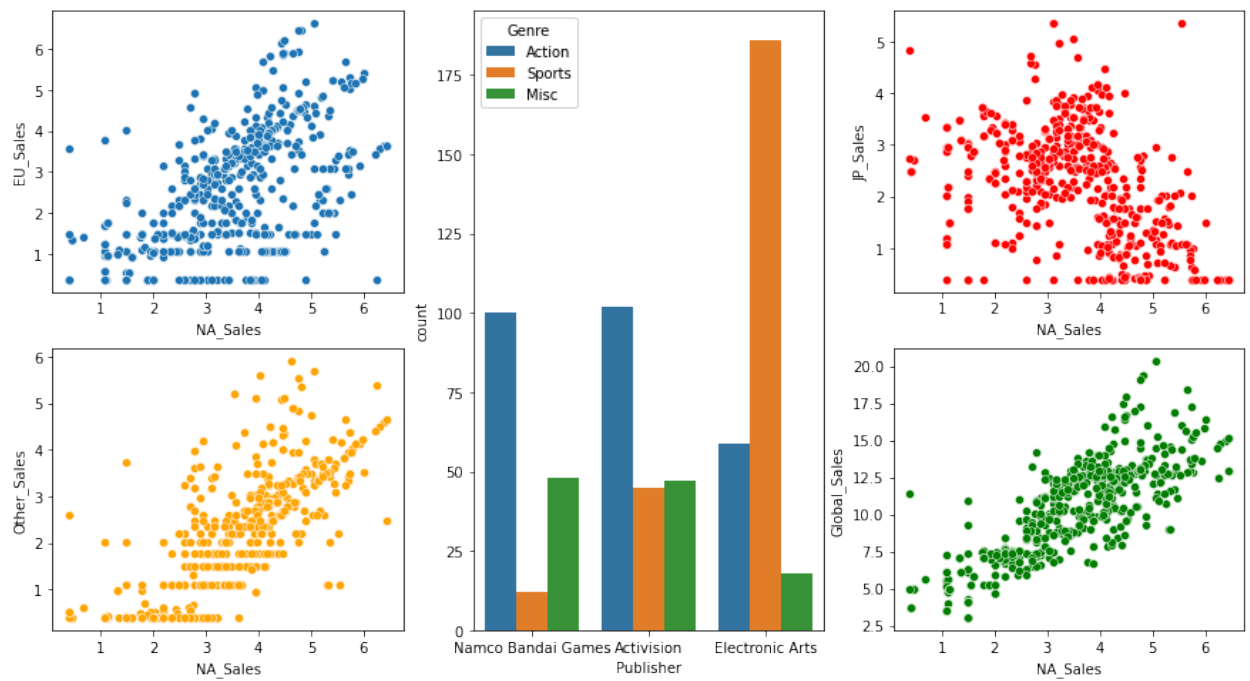
```
In [18]: plt.figure(figsize=(15,8))
plt.subplot(2, 3, 1)
sns.scatterplot(data = top3_data, x ='NA_Sales', y = 'EU_Sales')
plt.subplot(2, 3, 3)
sns.scatterplot(data = top3_data, x ='NA_Sales', y = 'JP_Sales', color = 'red')
plt.subplot(2, 3, 4)
sns.scatterplot(data = top3_data, x ='NA_Sales', y = 'Other_Sales',color='orange')
plt.subplot(2, 3, 6)
sns.scatterplot(data = top3_data, x ='NA_Sales', y = 'Global_Sales',color='green')
plt.subplot(1, 3 ,2)
sns.countplot(data = top3_data,x = 'Publisher', hue = 'Genre')
plt.suptitle("NA Sales Vs regions")
plt.show()
```

NA Sales Vs regions



```
In [21]: plt.figure(figsize=(15,8))
plt.subplot(2, 3, 1)
sns.scatterplot(data = top3_data, x = 'NA_Sales', y = 'EU_Sales')
plt.subplot(2, 3, 3)
sns.scatterplot(data = top3_data, x = 'NA_Sales', y = 'JP_Sales', color = 'red')
plt.subplot(2, 3, 4)
sns.scatterplot(data = top3_data, x = 'NA_Sales', y = 'Other_Sales', color='orange')
plt.subplot(2, 3, 6)
sns.scatterplot(data = top3_data, x = 'NA_Sales', y = 'Global_Sales', color='green')
plt.subplot(2, 3, 2)
sns.scatterplot(data = top3_data, x = 'NA_Sales', y = 'Global_Sales', color='green')
plt.subplot(1, 3, 2)
sns.countplot(data = top3_data, x = 'Publisher', hue = 'Genre')
plt.suptitle("NA Sales Vs regions")
plt.show()
```

NA Sales Vs regions



In []:

In []:

In []:

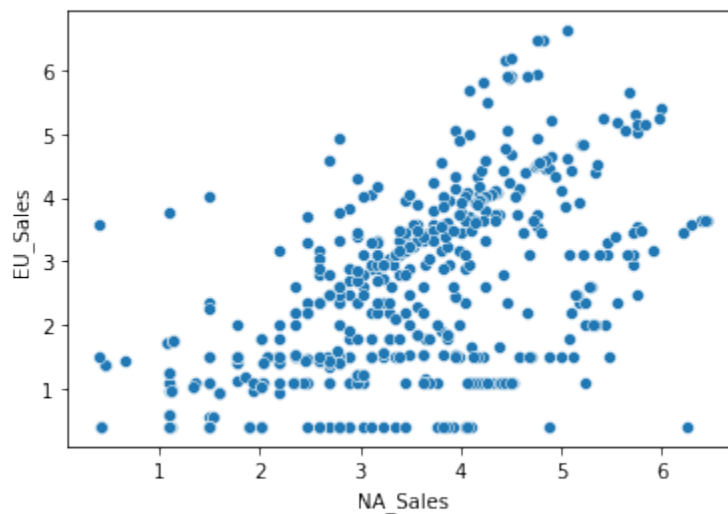
In []:

NNC

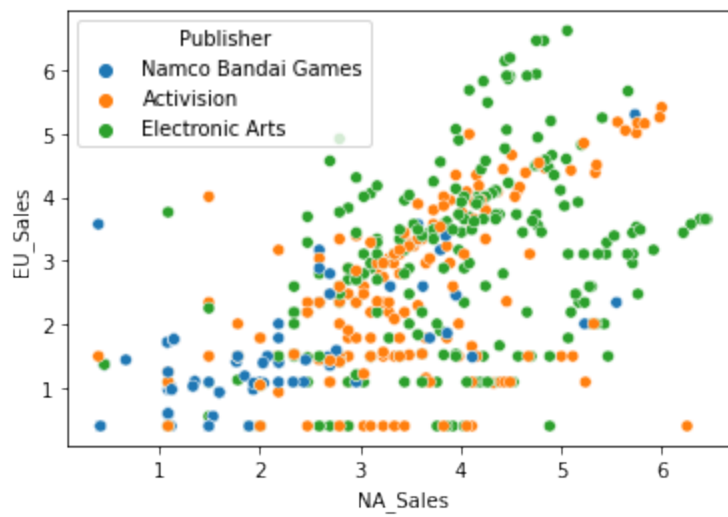
In [22]: `top3_data.head()`

Out[22]:	Rank	Name	Platform	Year	Genre	Publisher	NA_Sales	EU_Sales	J	
	2	14279	.hack: Sekai no Mukou ni + Versus	PS3	2012.0	Action	Namco Bandai Games	1.145709	1.762339	1
	13	2742	[Prototype 2]	PS3	2012.0	Action	Activision	3.978349	3.727034	0
	16	1604	[Prototype]	PS3	2009.0	Action	Activision	4.569217	4.108402	1
	19	1741	007: Quantum of Solace	PS3	2008.0	Action	Activision	4.156030	4.346074	1
	21	4501	007: Quantum of Solace	PS2	2008.0	Action	Activision	3.228043	2.738800	2

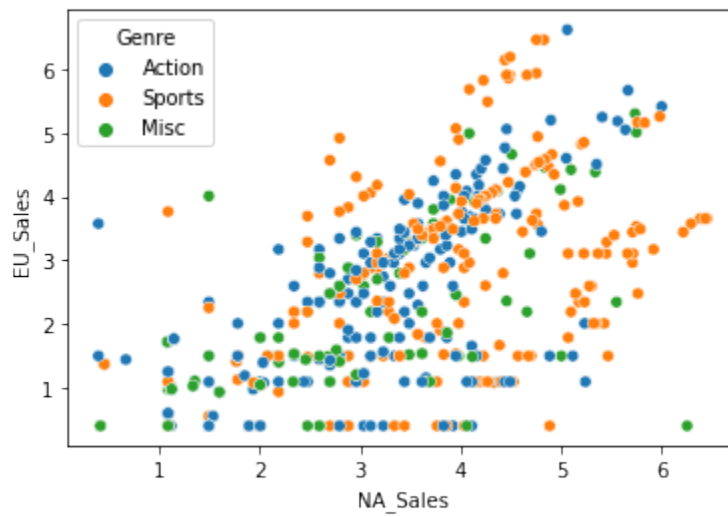
```
In [23]: sns.scatterplot(data = top3_data, x = 'NA_Sales', y = 'EU_Sales')
plt.show()
```



```
In [24]: sns.scatterplot(data = top3_data, x = 'NA_Sales', y = 'EU_Sales', hue = 'Publi')
plt.show()
```



```
In [25]: sns.scatterplot(data = top3_data, x = 'NA_Sales', y = 'EU_Sales', hue = 'Genre')
plt.show()
```



```
In [ ]:
```

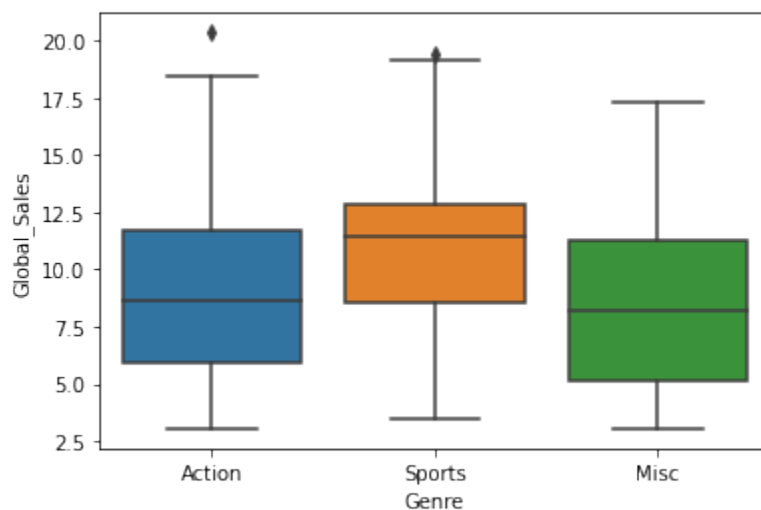
```
In [ ]:
```

CCN

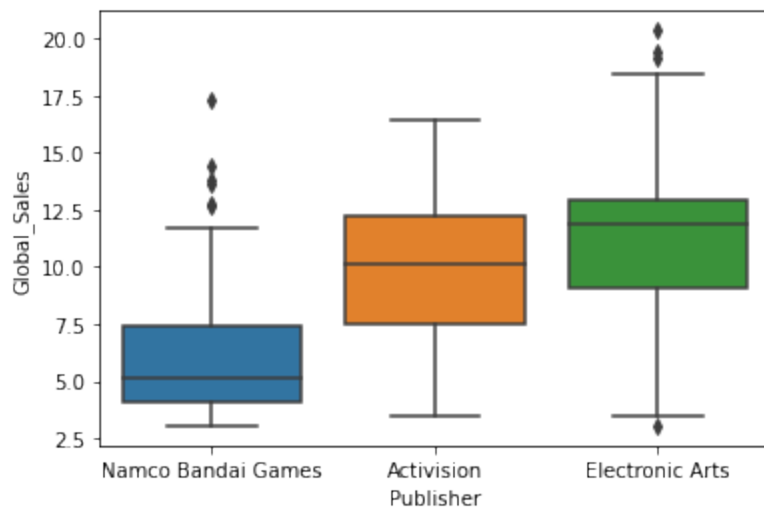
```
In [26]: top3_data.head()
```

Out[26]:	Rank	Name	Platform	Year	Genre	Publisher	NA_Sales	EU_Sales	J	
	2	14279	.hack: Sekai no Mukou ni + Versus	PS3	2012.0	Action	Namco Bandai Games	1.145709	1.762339	1
	13	2742	[Prototype 2]	PS3	2012.0	Action	Activision	3.978349	3.727034	0
	16	1604	[Prototype]	PS3	2009.0	Action	Activision	4.569217	4.108402	1
	19	1741	007: Quantum of Solace	PS3	2008.0	Action	Activision	4.156030	4.346074	1
	21	4501	007: Quantum of Solace	PS2	2008.0	Action	Activision	3.228043	2.738800	2

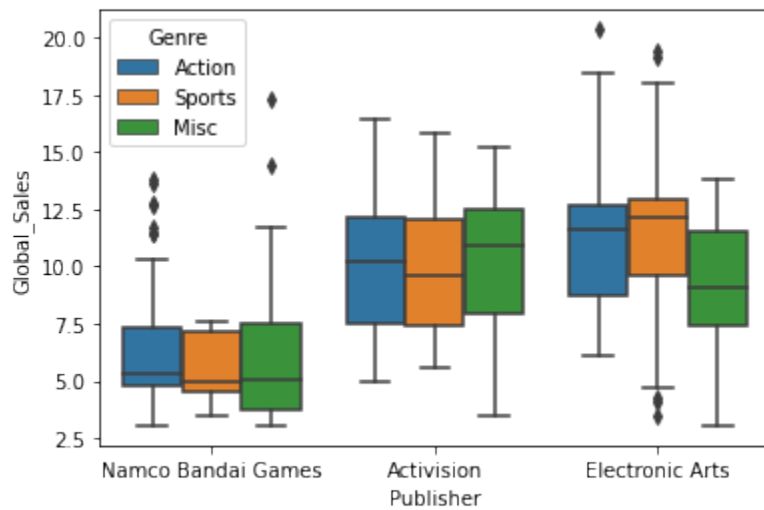
```
In [27]: sns.boxplot(data = top3_data, x = 'Genre', y = 'Global_Sales')
plt.show()
```



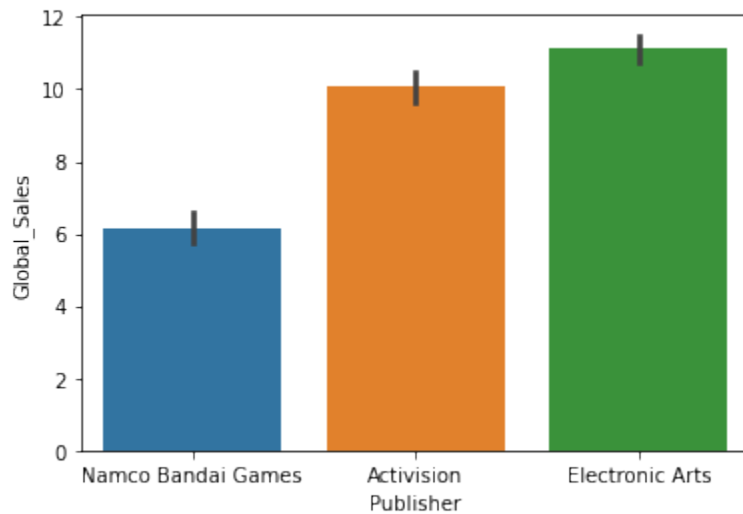
```
In [28]: sns.boxplot(data = top3_data, x = 'Publisher', y = 'Global_Sales')
plt.show()
```

```
In [29]: sns.boxplot(data = top3_data, x = 'Publisher', y = 'Global_Sales', hue = 'Genre')
plt.show()
```

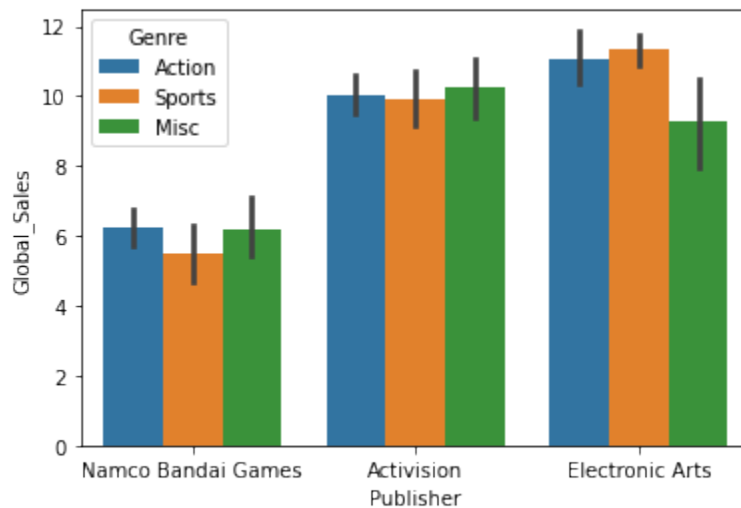


```
In [30]: sns.barplot(data = top3_data, x = 'Publisher', y = 'Global_Sales')
plt.show()
```



```
In [33]: # Dodged barplot
```

```
In [34]: sns.barplot(data = top3_data, x = 'Publisher', y = 'Global_Sales', hue = 'Genre',  
plt.show())
```



```
In [ ]:
```

```
In [ ]:
```

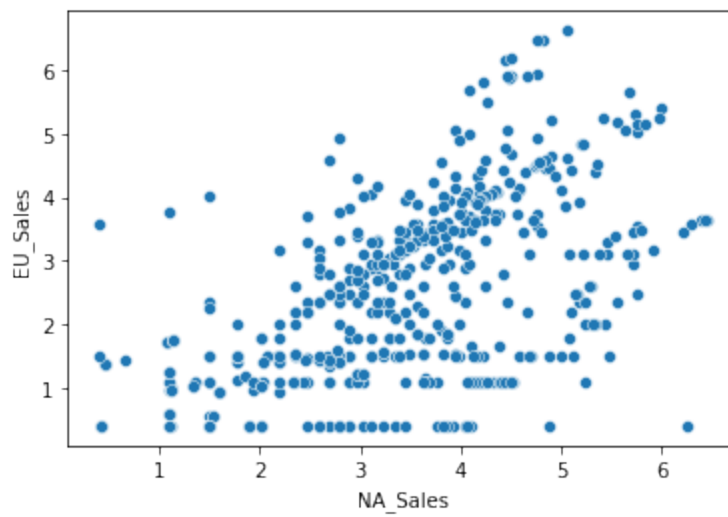
NNN

```
In [35]: top3_data.head()
```

```
Out[35]:
```

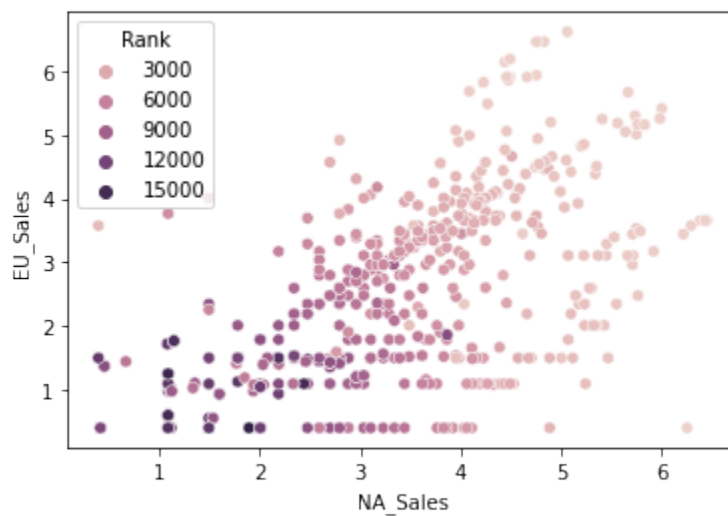
	Rank	Name	Platform	Year	Genre	Publisher	NA_Sales	EU_Sales	J	
	2	14279	.hack: Sekai no Mukou ni + Versus	PS3	2012.0	Action	Namco Bandai Games	1.145709	1.762339	1
	13	2742	[Prototype 2]	PS3	2012.0	Action	Activision	3.978349	3.727034	0
	16	1604	[Prototype]	PS3	2009.0	Action	Activision	4.569217	4.108402	1
	19	1741	007: Quantum of Solace	PS3	2008.0	Action	Activision	4.156030	4.346074	1
	21	4501	007: Quantum of Solace	PS2	2008.0	Action	Activision	3.228043	2.738800	2

```
In [36]: sns.scatterplot(data = top3_data, x = 'NA_Sales', y = 'EU_Sales')  
plt.show()
```

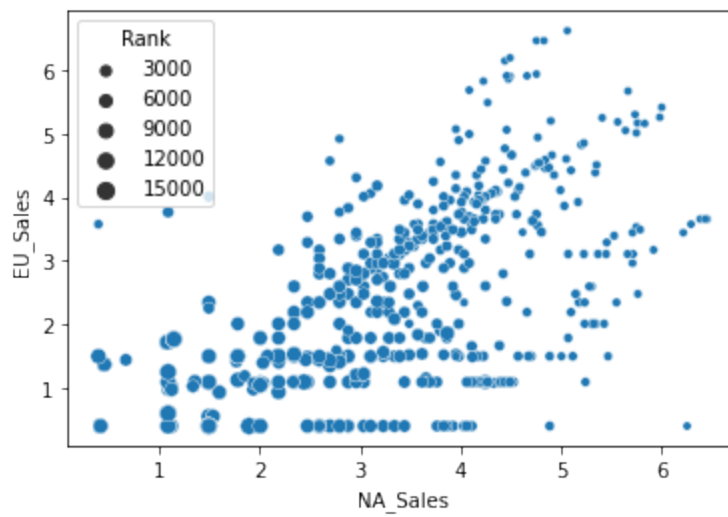


In []:

```
In [44]: sns.scatterplot(data = top3_data, x = 'NA_Sales', y = 'EU_Sales', hue = 'Rank',  
plt.show())
```



```
In [41]: sns.scatterplot(data = top3_data, x = 'NA_Sales', y = 'EU_Sales', size = 'Rank',  
plt.show())
```

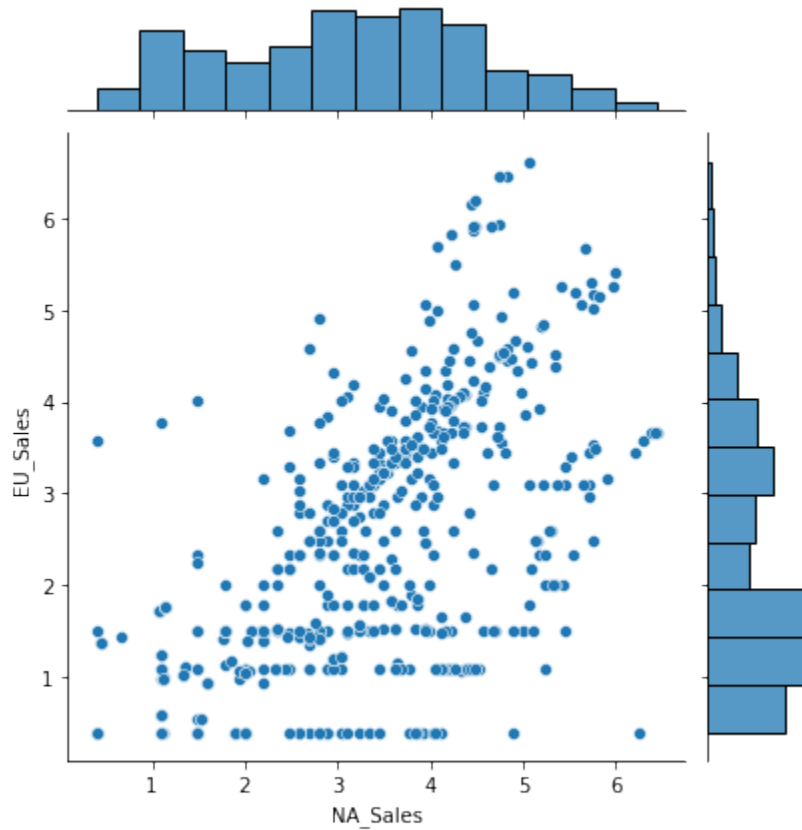


In []:

In []:

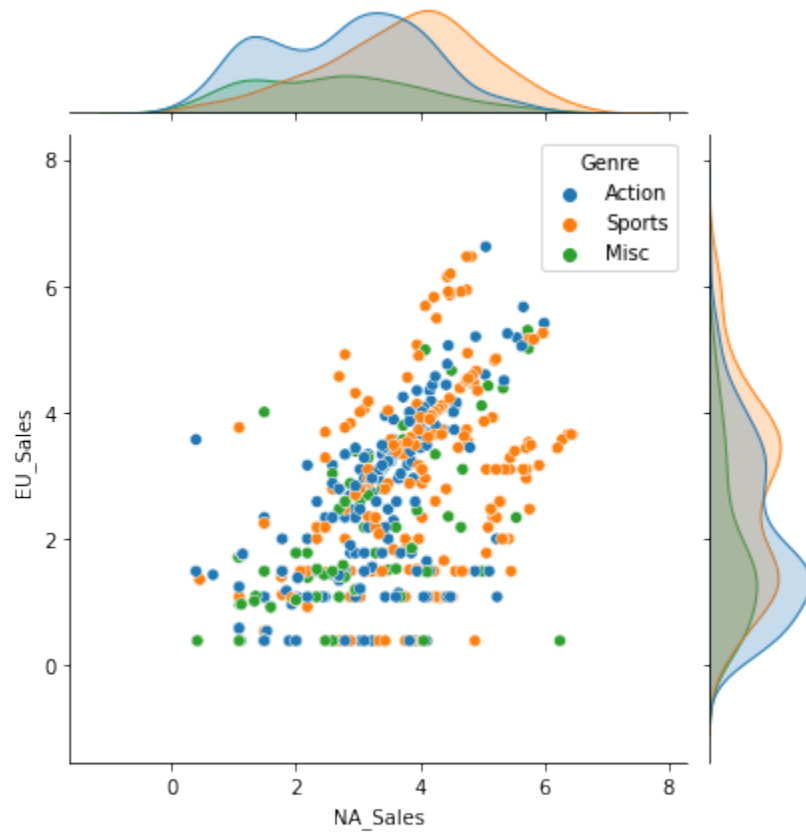
Joint_Plot

```
In [46]: sns.jointplot(data=top3_data, x = 'NA_Sales', y = 'EU_Sales')
plt.show()
```

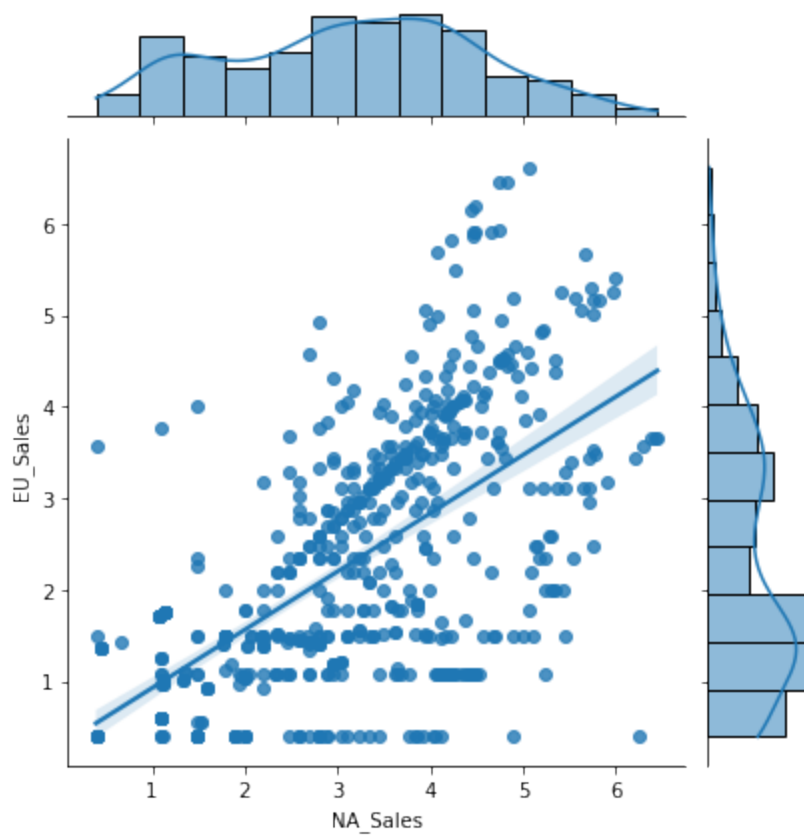


```
In [47]: sns.jointplot(data=top3_data, x = 'NA_Sales', y = 'EU_Sales', hue='Genre')
```

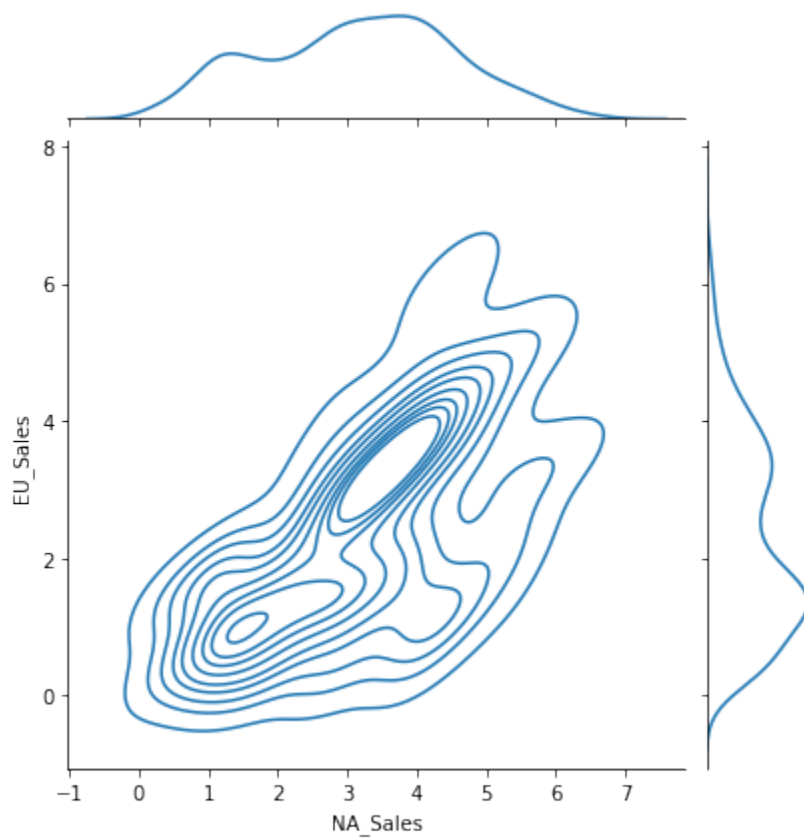
```
plt.show()
```



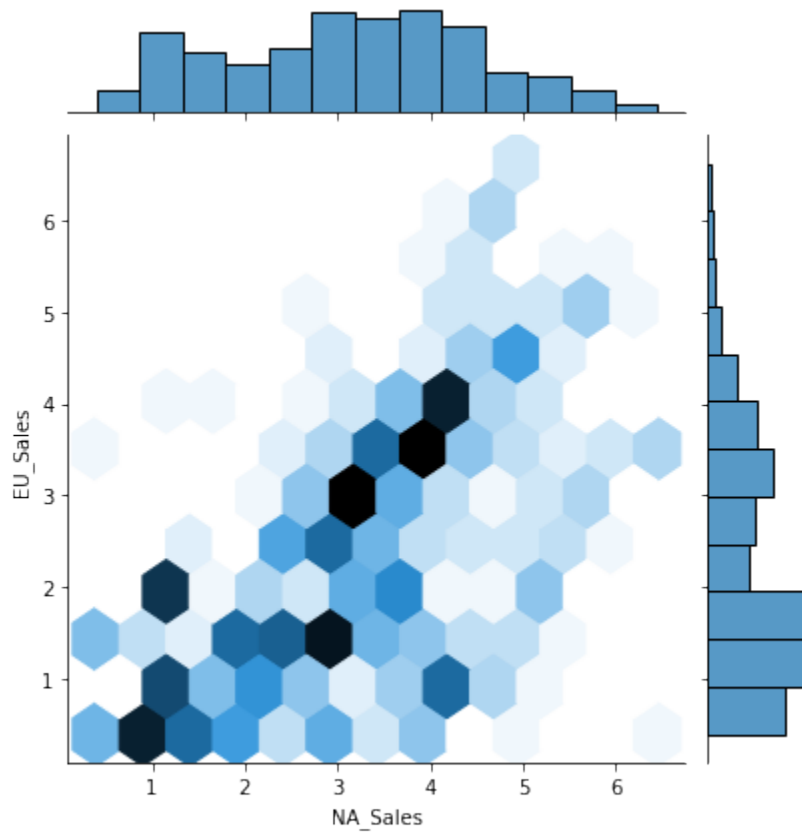
```
In [50]: sns.jointplot(data=top3_data, x = 'NA_Sales', y = 'EU_Sales', kind = 'reg')
plt.show()
```



```
In [51]: sns.jointplot(data=top3_data, x = 'NA_Sales', y = 'EU_Sales', kind = 'kde')
plt.show()
```



```
In [52]: sns.jointplot(data=top3_data, x = 'NA_Sales', y = 'EU_Sales', kind = 'hex')
plt.show()
```



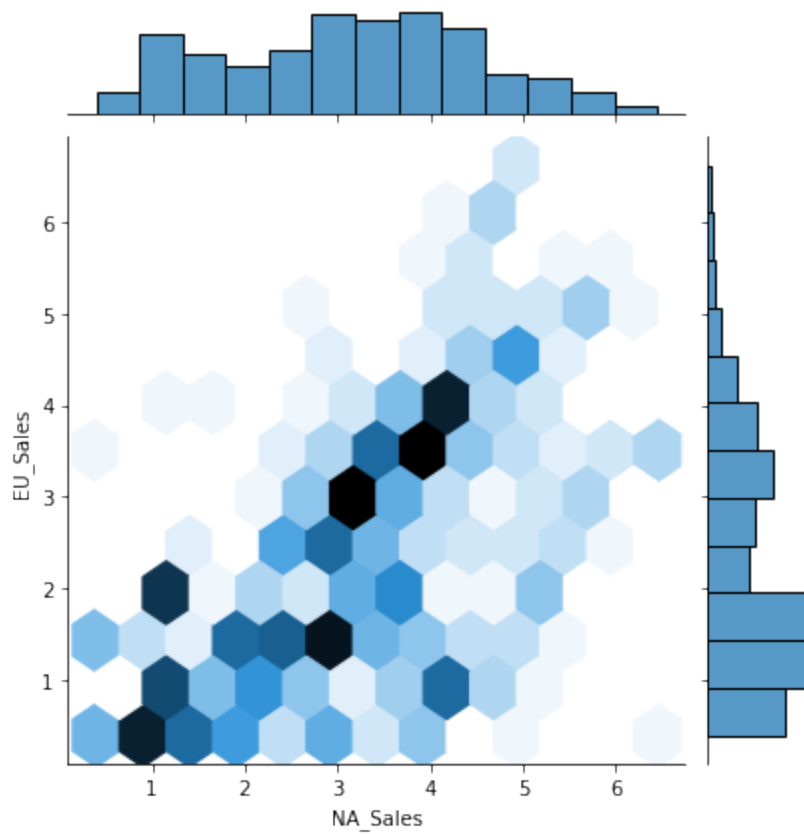
```
In [ ]:
```

```
In [ ]:
```

```
In [ ]:
```

Hexabin and heatmap

```
In [53]: sns.jointplot(data=top3_data, x = 'NA_Sales', y = 'EU_Sales', kind = 'hex')
plt.show()
```



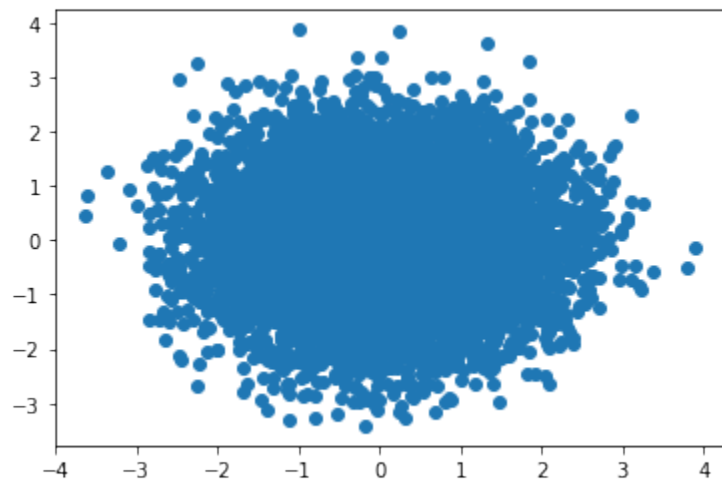
In []:

In [66]: `X = np.random.randn(10000, 2)`

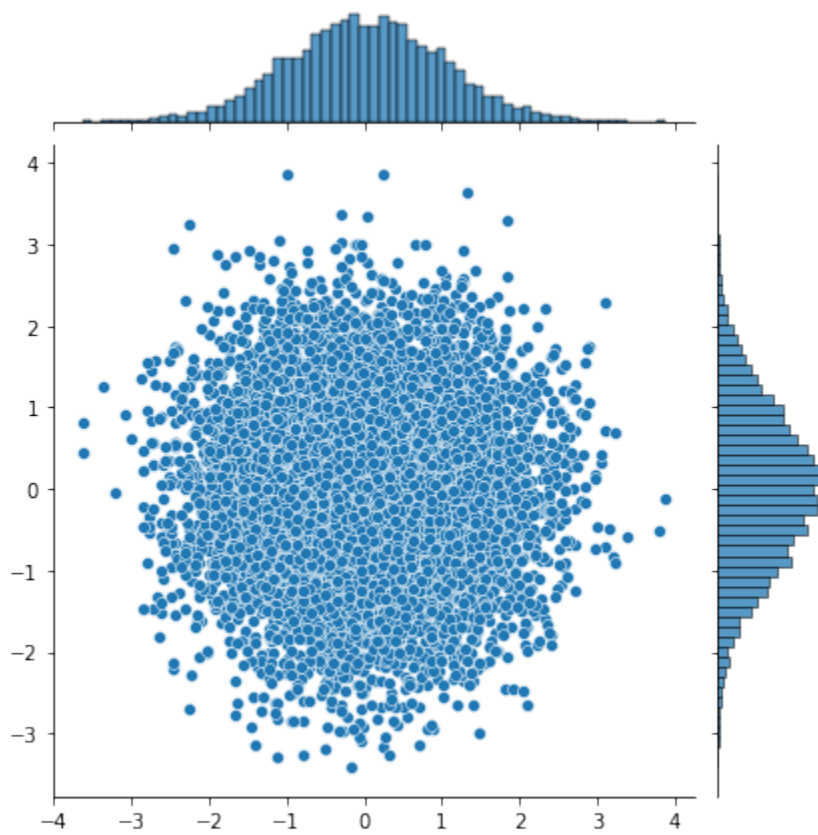
In [67]: `X.shape`

Out[67]: (10000, 2)

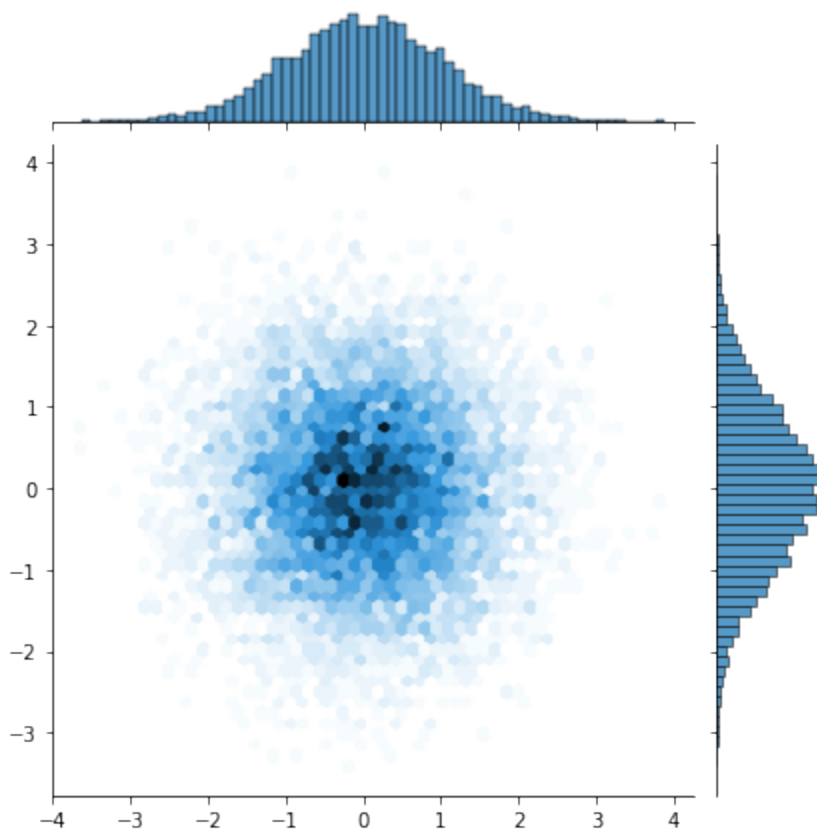
In [68]: `plt.scatter(X[:, 0], X[:, 1])`
`plt.show()`




```
In [69]: sns.jointplot(x = X[:, 0], y = X[:, 1])  
plt.show()
```



```
In [70]: sns.jointplot(x = X[:, 0], y = X[:, 1], kind = 'hex')  
plt.show()
```



In []:

In []:

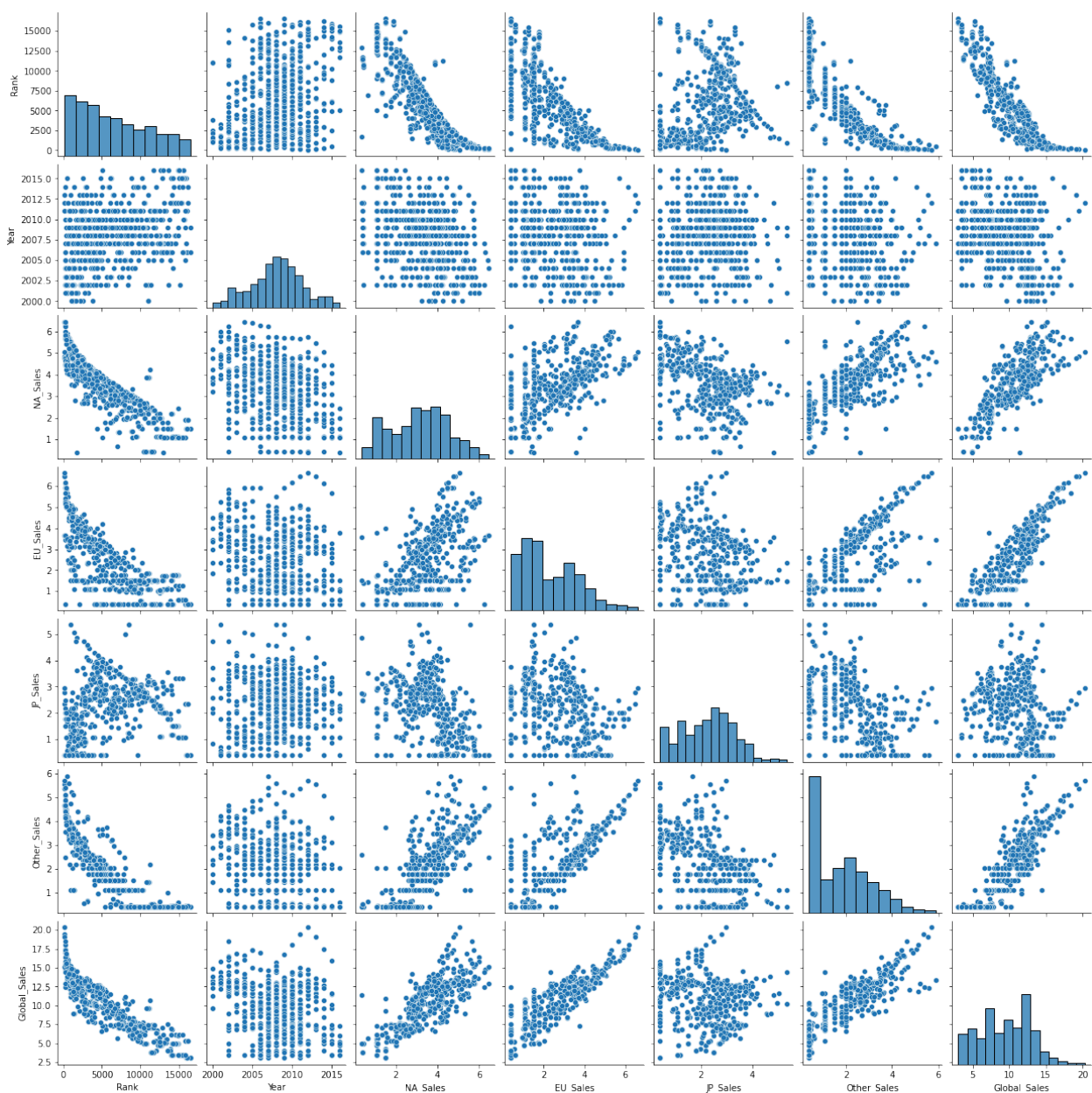
Pair-Plot

In [71]: `top3_data.head()`

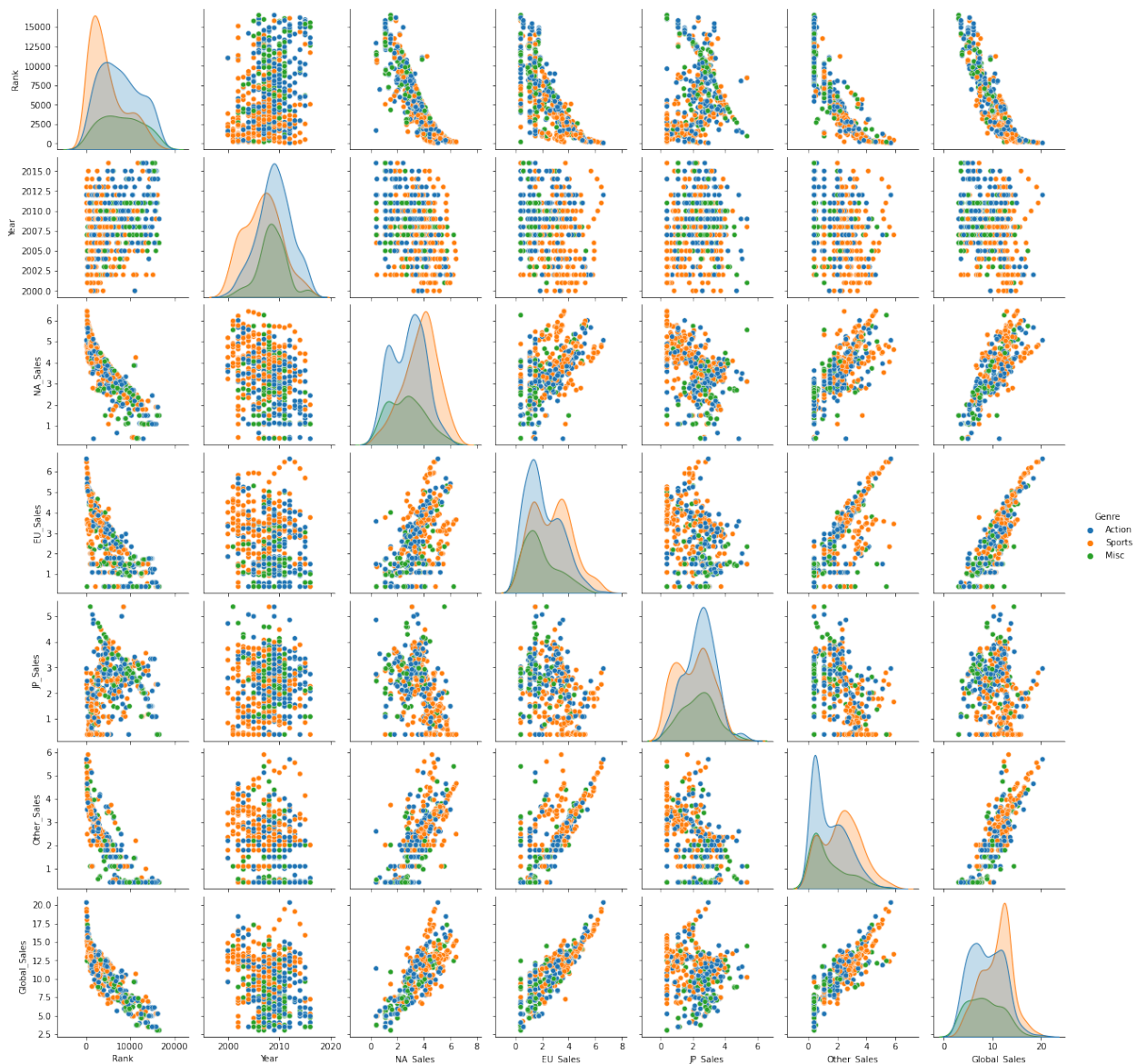
Out[71]:

	Rank	Name	Platform	Year	Genre	Publisher	NA_Sales	EU_Sales	J
2	14279	.hack: Sekai no Mukou ni + Versus	PS3	2012.0	Action	Namco Bandai Games	1.145709	1.762339	1
13	2742	[Prototype 2]	PS3	2012.0	Action	Activision	3.978349	3.727034	0
16	1604	[Prototype]	PS3	2009.0	Action	Activision	4.569217	4.108402	1
19	1741	007: Quantum of Solace	PS3	2008.0	Action	Activision	4.156030	4.346074	1
21	4501	007: Quantum of Solace	PS2	2008.0	Action	Activision	3.228043	2.738800	2

```
In [72]: sns.pairplot(data = top3_data)
plt.show()
```



```
In [73]: sns.pairplot(data = top3_data, hue='Genre')
plt.show()
```



In []:

In []:

Pearson Correlation

- Relation between data
- range: [-1, +1]

In [74]: `# pearson correlation`

In [75]: `top3_data.describe()`

Out[75]:

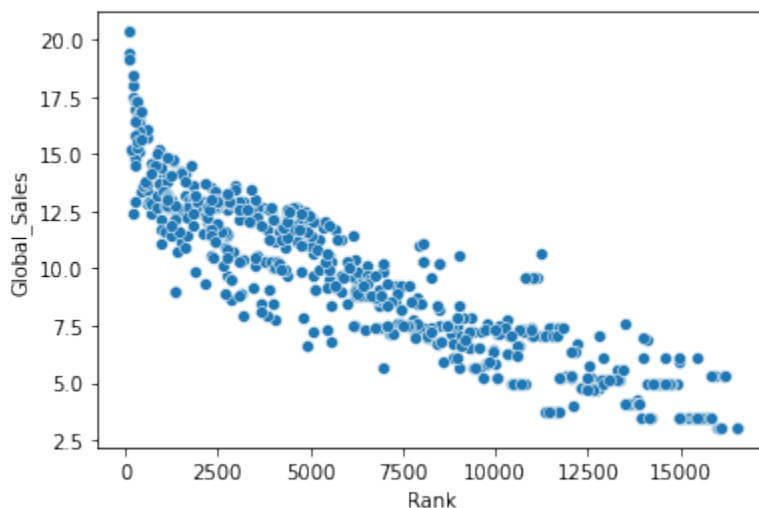
	Rank	Year	NA_Sales	EU_Sales	JP_Sales	Other_Sal
count	617.000000	611.000000	617.000000	617.000000	617.000000	617.0000
mean	6532.186386	2008.036007	3.145065	2.292522	2.264623	1.7865
std	4555.109759	3.496886	1.374738	1.416411	1.044837	1.3065
min	83.000000	2000.000000	0.394830	0.394830	0.394830	0.3948
25%	2609.000000	2006.000000	2.031986	1.087977	1.493442	0.3948
50%	5552.000000	2008.000000	3.228043	1.894355	2.353444	1.7811
75%	10041.000000	2010.000000	4.132499	3.390562	3.000569	2.6974
max	16532.000000	2016.000000	6.449269	6.619388	5.363208	5.9001

In [76]: `top3_data.corr()`

Out[76]:

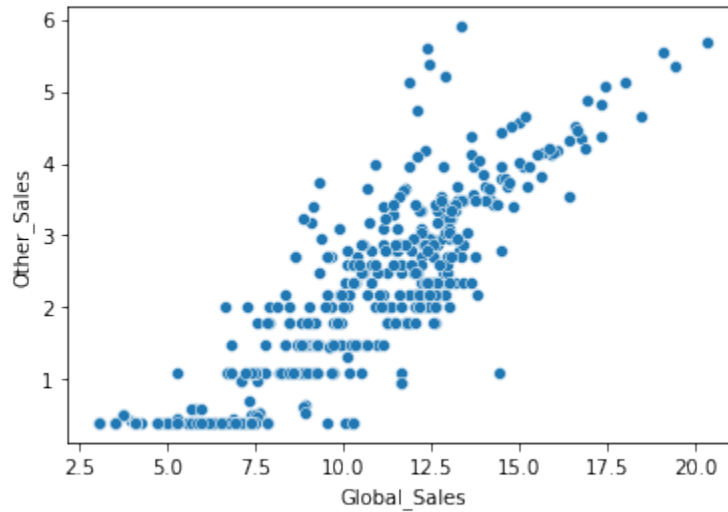
	Rank	Year	NA_Sales	EU_Sales	JP_Sales	Other_Sales	Global_Sales
Rank	1.000000	0.328705	-0.873726	-0.735711	0.115459	-0.857567	-0.911721
Year	0.328705	1.000000	-0.354256	-0.178026	0.055864	-0.239876	-0.280351
NA_Sales	-0.873726	-0.354256	1.000000	0.617483	-0.233315	0.794353	0.856300
EU_Sales	-0.735711	-0.178026	0.617483	1.000000	-0.208249	0.771105	0.864147
JP_Sales	0.115459	0.055864	-0.233315	-0.208249	1.000000	-0.355825	-0.014193
Other_Sales	-0.857567	-0.239876	0.794353	0.771105	-0.355825	1.000000	0.878816
Global_Sales	-0.911721	-0.280351	0.856300	0.864147	-0.014193	0.878816	1.000000

In [77]: `sns.scatterplot(data = top3_data, x = 'Rank', y = 'Global_Sales')`
`plt.show()`



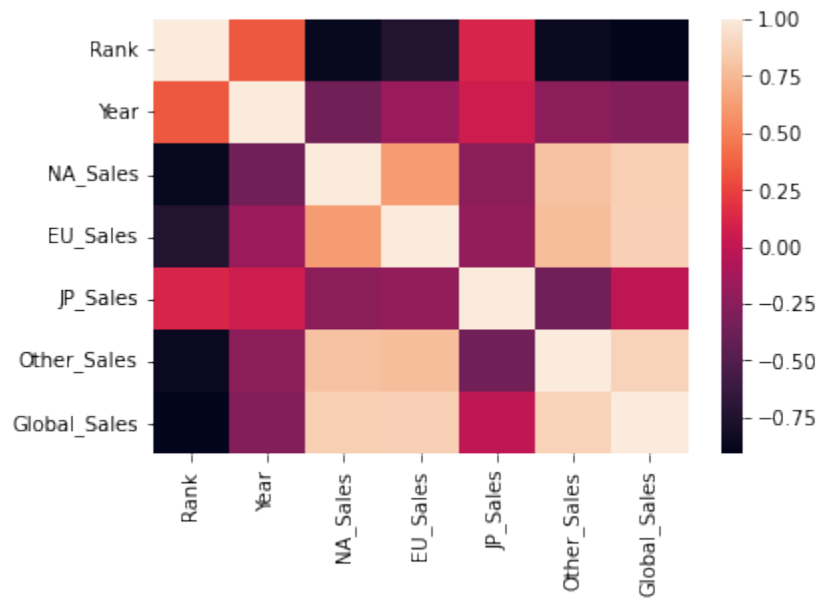
In [78]: `sns.scatterplot(data = top3_data, x = 'Global_Sales', y = 'Other_Sales')`

```
plt.show()
```

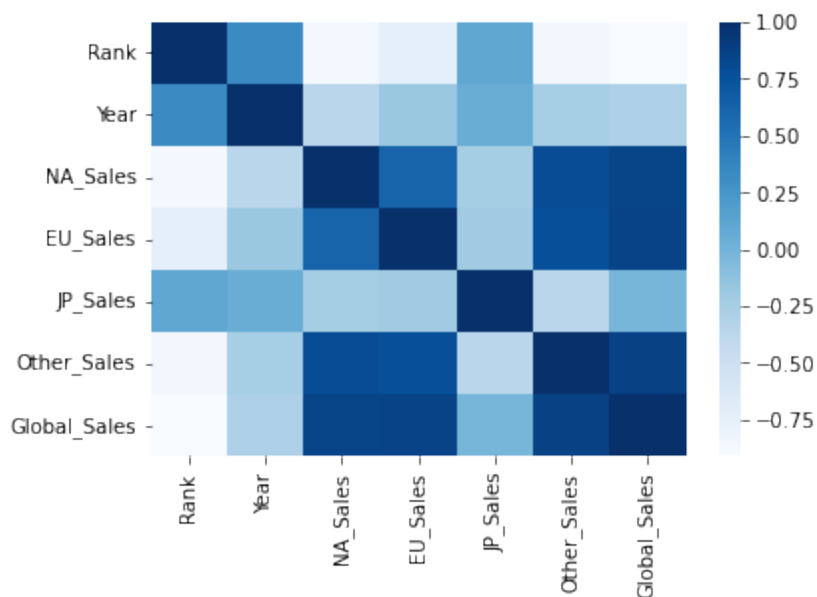


```
In [79]: corr = top3_data.corr()
```

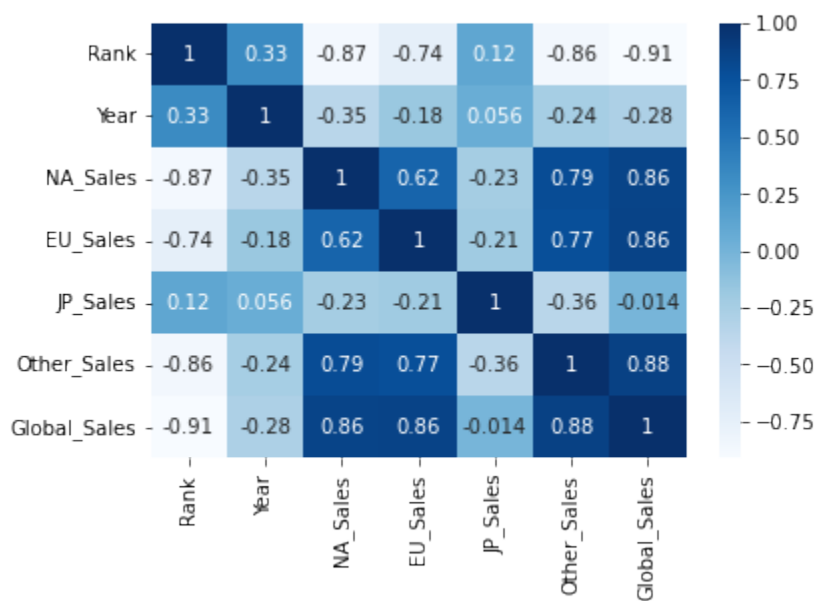
```
In [80]: sns.heatmap(corr)
plt.show()
```



```
In [85]: sns.heatmap(corr, cmap='Blues')
plt.show()
```



```
In [86]: sns.heatmap(corr, cmap='Blues', annot = True)
plt.show()
```



```
In [ ]:
```

```
In [ ]:
```

```
In [84]: print(plt.colormaps())
```

```
['Accent', 'Accent_r', 'Blues', 'Blues_r', 'BrBG', 'BrBG_r', 'BuGn', 'BuGn_r',  
'BuPu', 'BuPu_r', 'CMRmap', 'CMRmap_r', 'Dark2', 'Dark2_r', 'GnBu', 'GnBu_r',  
'Greens', 'Greens_r', 'Greys', 'Greys_r', 'OrRd', 'OrRd_r', 'Oranges', 'Orange  
s_r', 'PRGn', 'PRGn_r', 'Paired', 'Paired_r', 'Pastell', 'Pastell_r', 'Pastel  
2', 'Pastel2_r', 'PiYG', 'PiYG_r', 'PuBu', 'PuBuGn', 'PuBuGn_r', 'PuBu_r', 'PuO  
r_r', 'PuOr_r', 'PuRd', 'PuRd_r', 'Purples', 'Purples_r', 'RdBu', 'RdBu_r', 'RdG  
y', 'RdGy_r', 'RdPu', 'RdPu_r', 'RdYlBu', 'RdYlBu_r', 'RdYlGn', 'RdYlGn_r', 'Re  
ds', 'Reds_r', 'Set1', 'Set1_r', 'Set2', 'Set2_r', 'Set3', 'Set3_r', 'Spectra  
l', 'Spectral_r', 'Wistia', 'Wistia_r', 'YlGn', 'YlGnBu', 'YlGnBu_r', 'YlGn_r',  
'YlOrBr', 'YlOrBr_r', 'YlOrRd', 'YlOrRd_r', 'afmhot', 'afmhot_r', 'autumn', 'au  
tumn_r', 'binary', 'binary_r', 'bone', 'bone_r', 'brg', 'brg_r', 'bwr', 'bw  
r_r', 'cividis', 'cividis_r', 'cool', 'cool_r', 'coolwarm', 'coolwarm_r', 'copp  
er', 'copper_r', 'crest', 'crest_r', 'cubehelix', 'cubehelix_r', 'flag', 'fla  
g_r', 'flare', 'flare_r', 'gist_earth', 'gist_earth_r', 'gist_gray', 'gist_g  
ray_r', 'gist_heat', 'gist_heat_r', 'gist_ncar', 'gist_ncar_r', 'gist_rainbow',  
'gist_rainbow_r', 'gist_stern', 'gist_stern_r', 'gist_yarg', 'gist_yarg_r', 'gn  
uplot', 'gnuplot2', 'gnuplot2_r', 'gnuplot_r', 'gray', 'gray_r', 'hot', 'ho  
t_r', 'hsv', 'hsv_r', 'icefire', 'icefire_r', 'inferno', 'inferno_r', 'jet', 'j  
et_r', 'magma', 'magma_r', 'mako', 'mako_r', 'nipy_spectral', 'nipy_spectra  
l_r', 'ocean', 'ocean_r', 'pink', 'pink_r', 'plasma', 'plasma_r', 'prism', 'pri  
sm_r', 'rainbow', 'rainbow_r', 'rocket', 'rocket_r', 'seismic', 'seismic_r', 's  
pring', 'spring_r', 'summer', 'summer_r', 'tab10', 'tab10_r', 'tab20', 'tab2  
0_r', 'tab20b', 'tab20b_r', 'tab20c', 'tab20c_r', 'terrain', 'terrain_r', 'turb  
o', 'turbo_r', 'twilight', 'twilight_r', 'twilight_shifted', 'twilight_shifte  
d_r', 'viridis', 'viridis_r', 'vlag', 'vlag_r', 'winter', 'winter_r']
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